Examples

```
std::ratio_add<std::ratio<1,3>, std::ratio<2,5> >::num == 11
std::ratio_add<std::ratio<1,3>, std::ratio<2,5> >::den == 15

std::ratio_add<std::ratio<1,3>, std::ratio<7,6> >::num == 3
std::ratio_add<std::ratio<1,3>, std::ratio<7,6> >::den == 2
```

D.6.3. std::ratio_subtract template alias

The `std::ratio_subtract` template alias provides a mechanism for subtracting two `std::ratio` values at compile time, using rational arithmetic.

Definition

```
template <class R1, class R2>
using ratio_subtract = std::ratio<see below>;
```

Preconditions

`R1` and `R2` must be instantiations of the `std::ratio` class template.

Effects

`ratio_subtract<R1,R2>` is defined as an alias for an instantiation of `std::ratio` that represents the difference of the fractions represented by `R1` and `R2` if that difference can be calculated without overflow. If the calculation of the result overflows, the program is ill formed. In the absence of arithmetic overflow, `std::ratio_subtract<R1,R2>` shall have the same `num` and `den` values as `std::ratio<R1::num * R2::den - R2::num * R1::den, R1::den * R2::den>`.

Examples

```
std::ratio_subtract<std::ratio<1,3>, std::ratio<1,5> >::num == 2
std::ratio_subtract<std::ratio<1,3>, std::ratio<1,5> >::den == 15
```